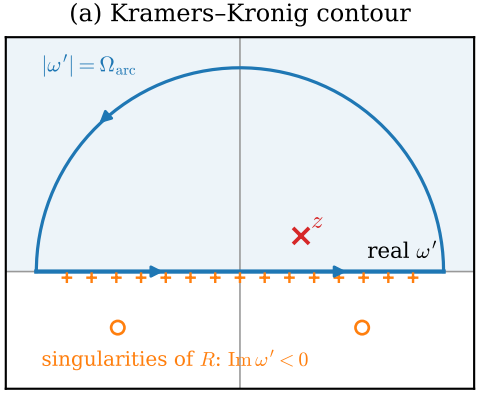
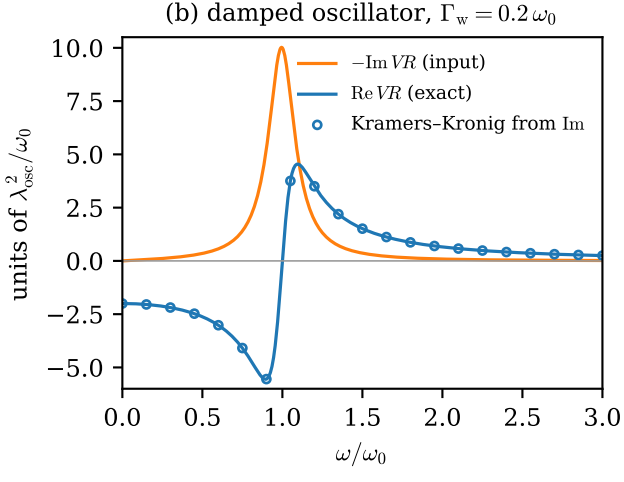

(a) Kramers–Kronig contour
$|\omega'| = \Omega_{\mathrm{arc}}$
z
real ω'
singularities of R: $\mathrm{Im}\,\omega' < 0$


(b) damped oscillator, $\Gamma_{\mathrm{w}} = 0.2\,\omega_0$
$-\mathrm{Im}\,VR$ (input)
$\mathrm{Re}\,VR$ (exact)
Kramers–Kronig from Im
units of $\lambda^2_{\mathrm{osc}}/\omega_0$
ω/ω_0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0